

GLASGOW COLLEGE UESTC

Exam paper

Electromagnetic Field and Microwave Technology (4002)

Date: (remember to complete when info available from Ruoli)

Time: (remember to complete when info available from Ruoli)

Attempt all PARTS. Total 100 marks

Use one answer sheet for each of the questions in this exam.

Show all work on the answer sheet.

For Multiple Choice Questions, use the dedicated answer sheet provided.

Make sure that your University of Glasgow and UESTC Student Identification Numbers are on all answer sheets.

An electronic calculator may be used provided that it does not allow text storage or display, or graphical display.

All graphs should be clearly labelled and sufficiently large so that all elements are easy to read.

The numbers in square brackets in the right-hand margin indicate the marks allotted to the part of the question against which the mark is shown. These marks are for guidance only.

DATA/FORMULAE SHEET IS PROVIDED AT THE END OF PAPER

- Q1 (a) What is the polarization of a dielectric in an electrostatic field? Differentiate the polarized charge in a dielectric from the conducting charge in a conducting material. [5]
- (b) Explain the Coulomb gauge and the Lorentz gauge; what is the proper situation to apply each of them? [6]
- (c) **Figure Q1 (a)** shows the cross-section view of an infinitely long rectangular waveguide; describe the equation and boundary condition of the electric potential inside the waveguide. [4]
- (d) Consider the coaxial line shown in **Figure Q1(b)**, with outer radius r_1 and inner radius r_2 . The region $r_2 < r < r_3$ is filled by dielectric 1 with permittivity ϵ_1 , and the region $r_3 < r < r_1$ is filled by dielectric 2 with permittivity ϵ_2 . Find the capacitance per unit length of the coaxial line? [10]

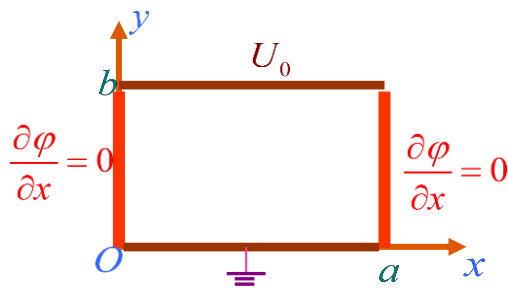


Figure Q1 (a)

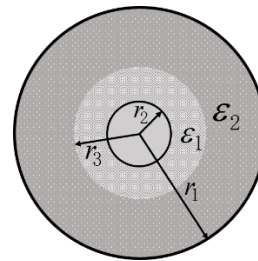


Figure Q1 (b)

[3]

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- Q2 (a) The interface for two media is the xoz plane, the magnetic fields on the interface are $\vec{H}_1 = 3\vec{e}_x + 5\vec{e}_y + \vec{e}_z$ ($y = 0_+$) and $\vec{H}_2 = -2\vec{e}_x + 3\vec{e}_y + 2\vec{e}_z$ ($y = 0_-$), what is the surface conducting current density $\vec{J}_s$ on the interface. If there are air for $y < 0$ and a material with $\mu_r = 3$ for $y > 0$, determine the magnetized surface current density $\vec{J}_{sm}$ on the interface? [6]
- (b) A coaxial line with inner radius a and outer radius b , is filled with air. If the inner radius change to $2a$ and outer radius is kept the same, what is the change for the internal self-inductance L_i and the external self-inductance L_o per unit length of the coaxial line; would it increase, decrease or stay the same? Explain the reason. [4]
- (c) A long straight wire I and a loop are placed inside the open space as shown in **Figure Q2**, the wire is at the interface of air (μ_0) and magnetic material (μ) where loop is located.
- (i) Find the magnetic field intensity $\vec{B}$ and the magnetic flux density $\vec{H}$. [6]
- (ii) Find the magnetized surface current density and magnetized line current. [6]
- (iii) Determine mutual inductance between the wire and the loop. [6]

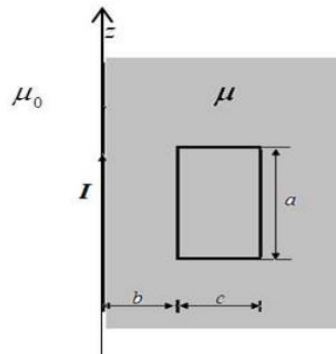


Figure Q2

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Q3 A uniform plane wave is propagating in a lossless medium with $\epsilon_r = 8$ and $\mu_r = 2$. $\vec{E}(x, y, z) = \left[(1 + mj)\vec{e}_x + (\sqrt{3} - j)\vec{e}_y \right] e^{-jkz}$ is the phasor form of the electric field intensity, where k is the wave number.

- (a) Describe the three polarization types of a uniform plane wave. [4]
- (b) Explain how to judge the polarization type of a uniform plane wave. [6]
- (c) If $m = \sqrt{3}$, what is the polarization type of this uniform plane wave. [6]
- (d) If $k = 2\pi (m^{-1})$, find the angular frequency of the uniform plane wave. [4]

Q4 A uniform plane wave is propagating in free space. Its electric field intensity is $\vec{E} = \vec{e}_x 240\pi e^{-j2\pi(y-z)}$.

- (a) Find the wavenumber, frequency, magnetic field intensity, and the average Poynting vector of the uniform plane wave. [8]
- (b) If there is an interface between dielectric and free space, oriented as shown in **Figure Q4**, solve the incident angle of the uniform plane wave. [5]
- (c) If the dielectric is perfect with $\epsilon_1 = 4\epsilon_0$, find the condition for the permeability μ_1 that allows for the uniform plane wave to transmit into the dielectric space with no reflection. Give one more example of ϵ_1, μ_1 for no reflection? [6]
- (d) If the dielectric is perfect with $\epsilon_1 = 8\epsilon_0$ $\mu_1 = 2\mu_0$, and the plane wave is incident on the interface, determine the reflection coefficient and transmission coefficient, and the total electric field intensity in free space $\vec{E}_1$. [8]

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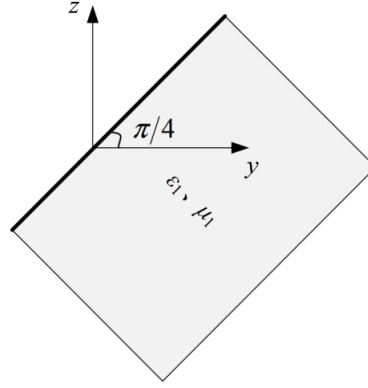


Figure Q4.

Appendix

Cylindrical coordinate system

$$\nabla \cdot \mathbf{A} = \frac{1}{\rho} \frac{\partial}{\partial \rho} (\rho A_\rho) + \frac{1}{\rho} \frac{\partial A_\phi}{\partial \phi} + \frac{\partial A_z}{\partial z}$$

$$\nabla \times \mathbf{A} = \frac{1}{\rho} \begin{vmatrix} \mathbf{e}_\rho & \rho \mathbf{e}_\phi & \mathbf{e}_z \\ \frac{\partial}{\partial \rho} & \frac{\partial}{\partial \phi} & \frac{\partial}{\partial z} \\ A_\rho & \rho A_\phi & A_z \end{vmatrix}$$

$$\nabla^2 u = \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial u}{\partial \rho} \right) + \frac{1}{\rho^2} \frac{\partial^2 u}{\partial \phi^2} + \frac{\partial^2 u}{\partial z^2}$$

Spherical coordinate system

$$\nabla \cdot \mathbf{A} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 A_r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta A_\theta) + \frac{1}{r \sin \theta} \frac{\partial A_\phi}{\partial \phi}$$

$$\nabla \times \mathbf{A} = \frac{1}{r^2 \sin \theta} \begin{vmatrix} \mathbf{e}_r & r \mathbf{e}_\theta & r \sin \theta \mathbf{e}_\phi \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial \phi} \\ A_r & r A_\theta & r \sin \theta A_\phi \end{vmatrix}$$

$$\nabla^2 u = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial u}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial u}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 u}{\partial \phi^2}$$

End of question paper